Case Study Report

Design and Creative Technologies

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Cloud Services and Deployment Models: A Cross-Sector Case Study Analysis

1. Introduction

Cloud computing has fundamentally reshaped how organizations across every sector deliver services, manage infrastructure, and drive automation. Rather than owning and operating physical data centers, enterprises now consume compute, storage, networking, and intelligence on demand – scaling elasticity, paying only for what they use, and offloading infrastructure management to specialist providers (Mell & Grance, 2011). The National Institute of Standards and Technology (NIST) defines cloud computing as a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort (Mell & Grance, 2011). The shift from Capital Expenditure (CAPEX)-heavy on-premises environments to Operational Expenditure (OPEX)-driven cloud models is no longer a trend exclusive to technology companies, it is a cross-industry imperative that spans government agencies, AI-native startups, and century-old industrial manufacturers alike (McHaney, 2021).

This report analyses three real-world case studies drawn from distinct sectors – government law enforcement (Victoria Police), artificial intelligence and software startup (ChatAndBuild), and industrial manufacturing (Voith) – each provided by the subject facilitator and representing a different combination of cloud service model, deployment model, and provider ecosystem. By examining how cloud was applied in each context, this report aims to: describe the essential cloud elements that enabled business automation in each case; distinguish cloud adoption from

traditional IT infrastructure approaches and identify the key cloud service offerings that made each outcome possible.

The three cases collectively illustrate how cloud computing is not a single, monolithic technology but a spectrum of services and deployment strategies calibrated to organizational context, risk appetite, and automation goals. Victoria Police leveraged Microsoft Power Platform (PaaS) within a hybrid multi-environment model to democratize application development across a law enforcement agency – transforming previously invisible, ungoverned Microsoft Access databases into a governed, scalable innovation ecosystem (Microsoft, 2026). ChatAndBuild, an AI-native startup based in Singapore, migrated to Google Cloud Platform to eliminate the manual infrastructure burden that was impeding real-time AI agent creation, achieving a 45% reduction in infrastructure overhead and a 30-70% improvement in generation latency (Google Cloud, n.d.). Voith, a 150-year-old manufacturing company operating across 60 countries, selected AWS following a competitive provider assessment to consolidate 3,500 servers distributed across 146 global locations into six AWS regions – projecting 30% cost savings compared to the legacy setup (Amazon Web Services, 2026).

2. Case Study 1 – Victoria Police & Microsoft Power Platform

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2.1. Case Overview

Victoria Police – Australia's principal state law enforcement agency – adopted Microsoft Power Platform to establish a **fusion development framework**: a governed model that

enables non-professional developers (police personnel across business units) to build data-driven applications safely, without depending solely on a constrained central IT team (Microsoft, 2026). The catalyst was a surge in self-directed application-building activity during the pandemic, when employees teaching themselves to use Power Apps revealed both the appetite for low-code tooling and the security risk of ungoverned 'shadow' applications built on standalone Microsoft Access databases. Rather than restricting this behavior, Victoria Police channeled it – creating structured pathways from personal productivity through to enterprise-grade production deployment, with two applications (Global Whereabouts and Bail Reporting) already progressing to enterprise adoption (Microsoft, 2026).

Figure 1 – ABC Enterprise cloud traffic flow - from user request through Route 53, load balancing, compute, storage, and database layers

2.2. Service Model Analysis

Victoria Police's cloud adoption aligns primarily with a **Platform as a Service (PaaS)** model. Power Apps and Power Platform abstract the underlying infrastructure – Microsoft manages the compute, networking, and database layers of Dataverse and the Power Platform runtime – while Victoria Police's developers focus on application logic, data models, and user experience (McHaney, 2021). This is the defining characteristic of PaaS: the provider controls the platform, the customer controls the application (IBM, n.d.0a). Some Microsoft 365 productivity workloads (SharePoint, Teams, Outlook) sit at the **SaaS** layer – fully managed software delivered over the network. There is no IaaS footprint in this case;

Victoria Police has not provisioned raw virtual machines or unmanaged storage – all development occurs within the abstracted Power Platform environment (Microsoft, 2026).

Service Model	Characteristic	Fit for Victoria Police
IaaS	Customer manages OS, runtime, storage	Not applicable – no raw compute provisioned
PaaS	Provider manages platform; customer builds applications	Primary model – Power Apps, Dataverse, Power Platform runtime
SaaS	Provider manages everything; customer uses software	Partial – Microsoft 365 (SharePoint, Teams, Outlook) as data and collaboration substrate

Table 1 – Service model analysis for Victoria Police (Microsoft, 2026).

2.3. Deployment Model Comparison

Victoria Police operates a hybrid, multi-environment deployment model: a personal productivity environment (the default Microsoft 365 tenant available to all staff), dedicated fusion environments (isolated, governed Power Platform environments provisioned on request by business units with executive endorsement), and enterprise production environments (reviewed and approved by the central Power Platform team) (Microsoft, 2026). This tiered model is a practical application of hybrid cloud architecture – multiple environment tiers serving different governance, security, and scale requirements simultaneously (Manvi & Shyam, 2021).

Deployment Model	Governance	Security	Scalability	Fit for Victoria Police
On-premises (Microsoft Access)	Low – invisible to IT	High risk – unadited shadow apps	None	Prior state – security blind spots, no transparency

Hybrid (Multi-env Power Platform)	High – tiered approval flow with exec endorsement	High – centralized standards, security review	Moderate	Adopted – balances innovation with enterprise controls
Public Cloud (ungoverned)	Low	Low	High	Creates same shadow IT risk as pre-migration state
Private Cloud	High	High	Limited	High cost, no low-code developer ecosystem available

Table 2 – Deployment model comparison for Victoria Police (Microsoft, 2026; Manvi & Shyam, 2021).

2.4. Cloud Services Analysis

The fusion framework relies on a layered set of Microsoft Power Platform services, each enabling a distinct stage of the application lifecycle (Microsoft, 2026):

Service	Purpose	Business Outcome
Power Apps (Low-code / Paas)	Enable non-professional developers to build data-driven apps without code	Personnel build and iterate solutions without the central IT bottleneck
Microsoft Dataverse	Model-driven, structure data store underpinning enterprise apps	Replaces isolated MS Access databases with a governed, transparent and auditable data layer
Power Platform Environments	Isolated development, test, and production workspaces per business unit	Clear pathway from personal innovation to enterprise-grade development
Power Platform Approval Flow	Governs requests for dedicated fusion environments via a model-driven app	Ensures executive endorsement and security team review before promotion to production

Microsoft 365 (SharePoint)	Data substrate and collaboration layer for initial prototypes	Familiar toolset lowers the barrier to entry for non-technical builders
Copilot Studio (exploration)	AI-assisted app development using generative coding	Future capability to further accelerate the build-to-deploy cycle

Table 3 – Microsoft Power Platform services utilised by Victoria Police and their business outcomes (Microsoft, 2026).

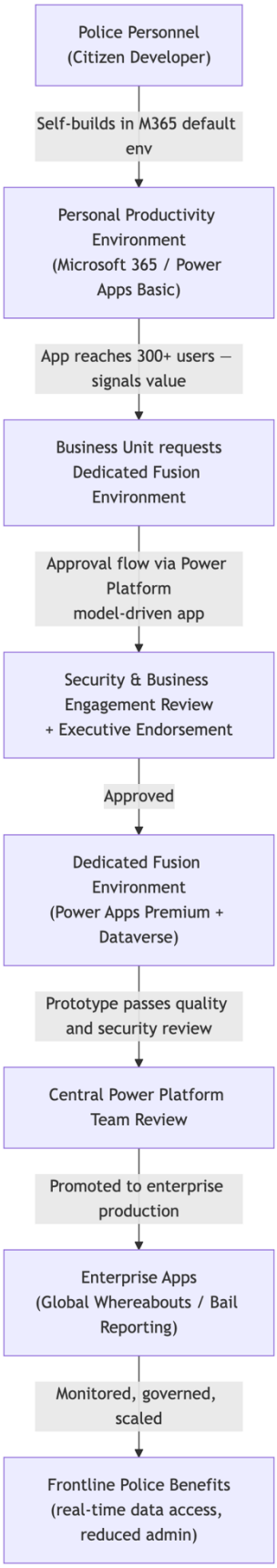
Police Personnel (Citizen Developer)
Self-builds in M365 default env
Personal Productivity Environment (Microsoft 365 / Power Apps Basic)
App reaches 300+ users — signals value
Business Unit requests Dedicated Fusion Environment
Approval flow via Power Platform model-driven app
Security & Business Engagement Review + Executive Endorsement
Approved
Dedicated Fusion Environment (Power Apps Premium + Dataverse)
Prototype passes quality and security review
Central Power Platform Team Review
Promoted to enterprise production
Enterprise Apps (Global Whereabouts / Bail Reporting)
Monitored, governed, scaled
Frontline Police Benefits (real-time data access, reduced admin)

Figure 1: Victoria Police fusion development lifecycle: from citizen developer prototype through tiered governance to enterprise production (Microsoft, 2026).

2.5. Reflection

The Victoria Police case demonstrates how cloud's **on-demand self-service** characteristic (Mell & Grance, 2011) can be safely extended to non-professional developers when paired with tiered governance. In a traditional IT model, every application request would queue behind the central IT team – a bottleneck that drove staff to build ungoverned Microsoft Access solutions outside the organization's visibility. The Power Platform model inverts this dynamic: rather than centralizing all development capacity, it governs the pathway to production while allowing distributed development (Microsoft, 2026). This represents a fundamental distinction from traditional IT: the cloud enables a spectrum of developer access that on-premises tooling structurally cannot support (McHaney, 2021).

However, risks remain significant. Sensitive law enforcement data held within a cloud-managed PaaS environment requires robust data residency controls and compliance with the Australian Protective Security Policy Framework (PSPF). As enterprise adoption scales to cover regions and thousands of officers, the risk surface expands correspondingly – a concern Efrain Tionko (Head of Digital Delivery and Transformation, Victoria Police) explicitly acknowledged when noting the need to increase controls as employee access grows (Microsoft, 2026).

3. Case Study 2 – ChatAndBuild & Google Cloud

Cloud adoption is not risk-free. Three challenges are most relevant for ABC (Eliaçık, 2022):

1. **Security and Privacy:** ABC handles payments and customer PII. Security is the top concern for cloud adopters - 90% of security professionals cite it as a challenge, alongside governance (71%) and compliance (68%) (Manvi & Shyam, 2021). Mitigation: least privilege IAM, mandatory MFA, encryption at rest/in transit, and continuous alerting - the shared responsibility model clarifies the provider/customer division (Shore, 2020).
2. **Cost Volatility:** Pay-as-you-go can spiral without guardrails - overprovisioned instances and excessive egress generate surprise bills (Bittok, 2022). **Mitigation:** FinOps habits from day one: budget alerts, resource tagging, rightsizing, and reserved pricing for stable workloads.
3. **Vendor lock-in and Skills gap:** Deeper managed-service adoption makes provider migration expensive (Eliaçık, 2022). Mitigation: prioritize portability (containers, standard databases) and invest in targeted upskilling (McHaney, 2021).

4. Case Study 3 – Voith & AWS

Cloud service models sit on a control-versus-responsibility spectrum (IBM, n.d.-a; Linthicum, 2021; McHaney, 2021). IaaS provides full compute flexibility; PaaS abstracts infrastructure so the team can focus on development; SaaS offers limited customisation - less suited to a start-up that must differentiate its platform.

Deployment models vary in elasticity and cost. Community cloud does not apply to ABC's consumer use case. Private cloud sacrifices elasticity; hybrid introduces complexity that exceeds ABC's current maturity (Manvi & Shyam, 2021). Public cloud is the most strategically aligned option: IBM (n.d.-b) reports IaaS workloads experience 60% fewer security incidents than traditional data centers (see Table 1).

Deployment Model	Cost	Elasticity	ABC Fit
Public Cloud	Low (OPEX only)	High / auto-scales	Recommended
Private Cloud	High (CAPEX + ops staff)	Limited / fixed capacity	Over-engineered for startup
Hybrid Cloud	Medium - dual infra	Moderate / complex to manage	Premature / exceeds current maturity

Table 1 – Deployment model trade-offs for ABC Enterprise

Recommendation: A blended IaaS/PaaS approach - IaaS for compute flexibility, PaaS for managed databases and serverless functions - preserves rapid elasticity and measured service (Mell & Grance, 2011). A Virtual Private Cloud (VPC) can address future network isolation needs (IBM, n.d.-c).

5. Conclusion

Cloud computing is the right strategic move for ABC: measured service, rapid elasticity, and resource pooling (Mell & Grance, 2011) translate directly into automation outcomes - scaling without procurement delays, zero headcount growth, faster time-to-market. Recommended stack:

public cloud, blended IaaS/PaaS, hybrid cost model, AWS. Managed well, ABC's infrastructure stops being a bottleneck and becomes a competitive advantage (Eliaçık, 2022).

6. Appendices

6.1. Appendix A – Cloud vs Traditional IT Comparison Table

Dimension	Traditional IT	Cloud (ABC Case)
Cost model	CAPEX heavy	OPEX elasticity
Scaling	Hardware procurement	Auto Scaling
Deployment speed	Weeks/months	Minutes
Automation	Manual ops	Policy-driven

Table 3 – Cloud vs Traditional IT Comparison.

6.2. Appendix G – Glossary of Terms

Term	Meaning
CAPEX	Capital Expenditure
OPEX	Operational Expenditure
IAM	Identity and Access Management
TCO	Total Cost of Ownership
SOA	Service-Oriented Architecture
PaaS	Platform as a Service
IaaS	Infrastructure as a Service
SaaS	Software as a Service
VPC	Virtual Private Cloud
ELB	Elastic Load Balancer
EC2	Elastic Compute Cloud
RDS	Relational Database Service
Route 53	AWS DNS Service
Auto Scaling	AWS service for automatic scaling of compute resources

Shared Responsibility	Cloud security model defining provider vs customer security obligations
Measured Service	Resource usage is monitored and billed accordingly
Rapid Elasticity	Allows resources to scale up/down automatically based on demand

Table 8 – Glossary of Terms

End of Appendix Section

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